

Guessing Game

Create a GUI for a simple mathematical guessing game. The GUI should include the following Widgets:

- A Button to start the Game (Start-Button)
- A Label to print out answers to the guesses (Answer-Label)
- A Label to print out the current guess (Guess_label)
- An entry field in which the user can type it's starting guess (Guess-Entry)
- Buttons for increasing/decreasing the value of the current guess by 1/5/10/25/50/100/250/1000 (16 in total, Change-Buttons)
- A Button to start the comparison between the guessed number and the random number (Guess-Button)
- A Label for the Win Message (Win-Label)

The game functions as follows. When the user clicks the Start-Button a random generator creates a number between 1 and 2000, and this number is then saved in an integer VarVar. Make sure that the Guess-Button is deactivated (use state="disabled" in the constructor) on the start of the GUI and only gets activated once the user clicks on the Start-Button (use `button.config(state="normal")`). The start button should be deactivated at the end of the function. Now the user can type in his starting guess in the entry field and compare the guess with the random number by clicking on the Guess-Button. If the two are identical adjust the Win-Label to display a multi-line Win Message. When the guess doesn't match tell the user whether he guessed too high or too low in the Answer-Label. Don't forget to display the guessed number in the Guess-Label.

If the user guessed wrong on the first try, deactivate the entry field and let the user adjust his guess with the help of the Change-Buttons. The adjusted guess should always be displayed on the Guess-Label. Make sure that the guess of the user stays within the allowed bounds of 1 to 2000. When he's confident in his guess he can click the Guess-Button again and a new comparison takes place.

Once the user guessed right a multi-line winning message should be displayed on the Win-label and the Start-Button as well as the Guess-Entry should be activated again. Don't forget to deactivate the Guess-Button here as well.

Optional

Make the game a bit more interesting by setting a maximum number of guesses for the user. Start with a maximum of 20 guesses and display the remaining guesses on a fourth Label (Check-Label). Later on let the user define how many guesses he wants to take at maximum by adding a second entry (Check-Entry) and read in the value from this Check-Entry once the user clicks on the Guess-Button for the first time.